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DIFFERENCE-IN-DIFFERENCES WITH VARIATION IN TREATMENT TIMING

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In contrast to our substantial understanding of the canonical 2x2 DD model, we know relatively little about the two-way fixed effects DD model when treatment timing varies. We do not know precisely how it compares mean outcomes across groups.⁵ We typically rely on general descriptions of the identifying assumption like “interventions must be as good as random, conditional on time and group fixed effects” (Bertrand, Duflo, and Mullainathan 2004, p. 250), and consequently lack well-defined strategies to test the validity of the DD design with timing. We have limited understanding of the treatment effect parameter that regression DD identifies. Finally, we often cannot evaluate when alternative specifications will work or why they change estimates.⁶

This paper shows that the two-way fixed effects DD estimator in (2) is a weighted average of all possible 2x2 DD estimators that compare timing groups to each other. Some use units treated at a particular time as the treatment group and untreated units as the control group. Some compare units treated at two different times, using the later group as a control before its treatment begins and then the earlier group as a control after its treatment begins. As in any least squares estimator, the weights on the 2x2 DD’s are proportional to group sizes *and* the variance of the treatment dummy within each pair. Treatment variance is highest for groups treated in the middle of the panel and lowest for groups treated at the extremes. This result clarifies the theoretical interpretation and identifying assumptions of the general DD model and creates simple new tools for describing the design and analyzing problems that arise in practice.

By decomposing the DD estimator into its sources of variation (the 2x2 DD’s) and providing an explicit interpretation of the weights in terms of treatment variances, my results

⁵ Imai, Kim, and Wang (2018) note “It is well known that the standard DiD estimator is numerically equivalent to the linear two-way fixed effects regression estimator if there are two time periods and the treatment is administered to some units only in the second time period. Unfortunately, this equivalence result does not generalize to the multi-period DiD design...Nevertheless, researchers often motivate the use of the two-way fixed effects estimator by referring to the DiD design (e.g., Angrist and Pischke, 2009).”

⁶ This often leads to sharp disagreements. See Neumark, Salas, and Wascher (2014) on unit-specific linear trends, Lee and Solon (2011) on weighting and outcome transformations, and Shore-Sheppard (2009) on age time fixed effects.

extend recent research on DD models with heterogeneous effects.⁷ Assuming equal counterfactual trends, Abraham and Sun (2018), Borusyak and Jaravel (2017), and de Chaisemartin and D’Haultfœuille (2018b) show that two-way fixed effects DD yields an average of treatment effects across all groups and times, some of which may have negative weights. My results show how these weights arise from differences in timing and thus treatment variances, facilitating a connection between models of treatment allocation and the interpretation of DD estimates.⁸ I also explain why the negative weights occur: when already-treated units act as controls, *changes* in their treatment effects over time get subtracted from the DD estimate. This negative weighting only arises when treatment effects vary over time, in which case it typically biases regression DD estimates away from the sign of the true treatment effect. This does not imply a failure of the underlying *design*, but it does caution against the use of a single-coefficient two-way fixed effects specification to summarize time-varying effects.

I also show that because regression DD uses group sizes and treatment variances to weight up simple estimates that each rely on common trends between two groups, its identifying assumption is a variance-weighted version of common trends between all groups. The extent to which a group’s differential trend biases the overall estimate equals the difference between how much weight it gets when it acts as the treatment group and how much weight it gets when it acts as the control group. When the earliest and/or latest treated units have low treatment variance, they can get *more* weight as controls than treatments. In designs without untreated units they always

⁷ Early research in this area made specific observations about stylized specifications such as models with no unit fixed effects (Bitler, Gelbach, and Hoynes 2003), or it provided simulation evidence (Meer and West 2013).

⁸ Related results on the weighting of heterogeneous treatment effects does not provide this intuition. Abraham and Sun (2018, p 9) describe the weights in a DD estimate with constant treatment effects as “residual[s] from predicting treatment status, $D_{i,t}$ with unit and time fixed effects.” de Chaisemartin and D’Haultfœuille (2018b, p 7) and Borusyak and Jaravel (2017) describe these same weights as coming from an auxiliary regression and Borusyak and Jaravel (2017, p 10-11) note that “a general characterization of [the weights] does not seem feasible.” Athey and Imbens (2018) also decompose the DD estimator and develop design-based inference methods for this setting. Strezhnev (2018) expresses $\hat{\beta}^{DD}$ as an unweighted average of DD-type terms across pairs of observations and periods.

balanced panel and partial out unit and time fixed effects from y_{it} and D_{it} using the Frisch-Waugh theorem (Frisch and Waugh 1933). Denote grand means by $\bar{\bar{x}} = \frac{1}{NT} \sum_i \sum_t x_{it}$, and adjusted variables by $\tilde{x}_{it} = (x_{it} - \bar{\bar{x}}) - (\bar{x}_i - \bar{\bar{x}}) - (\bar{x}_t - \bar{\bar{x}})$. $\hat{\beta}^{DD}$ then equals the univariate regression coefficient between adjusted outcome and treatment variables:

$$\frac{\widehat{cov}(\tilde{y}_{it}, \tilde{D}_{it})}{\widehat{var}(\tilde{D}_{it})} = \frac{\frac{1}{NT} \sum_i \sum_t \tilde{y}_{it} \tilde{D}_{it}}{\frac{1}{NT} \sum_i \sum_t \tilde{D}_{it}^2}$$

The numerator equals the sample covariance between y_{it} and D_{it} minus the sample covariances between unit means and between time means:

$$\widehat{cov}(\tilde{y}_{it}, \tilde{D}_{it}) = \frac{1}{NT} \sum_i \sum_t (y_{it} - \bar{y}) (D_{it} - \bar{D}) - \frac{1}{N} \sum_i (\bar{y}_i - \bar{y}) (\bar{D}_i - \bar{D}) - \frac{1}{T} \sum_t (\bar{y}_t - \bar{y}) (\bar{D}_t - \bar{D}) \quad (6)$$

The appendix shows how to simplify this covariance using the binary nature of D_{it} , and by replacing the $\bar{y}_i$ and $\bar{y}_t$ terms with weighted averages of pre- and post-treatment means or means in each group.¹⁰ This gives the following theorem:

Theorem 1. Difference-in-Differences Decomposition Theorem

Assume that the data contain $k = 1, \dots, K$ groups of units ordered by the time when they receive a binary treatment, $t_k^* \in (1, T]$. There may be one group, U , that never receives treatment. The OLS estimate, $\hat{\beta}^{DD}$, in a two-way fixed-effects model (2) is a weighted average of all possible two-by-two DD estimators.

$$\hat{\beta}^{DD} = \sum_{k \neq U} s_{kU} \hat{\beta}_{kU}^{2x2} + \sum_{k \neq U} \sum_{\ell > k} s_{k\ell} [\mu_{k\ell} \hat{\beta}_{k\ell}^{2x2,k} + (1 - \mu_{k\ell}) \hat{\beta}_{k\ell}^{2x2,\ell}] \quad (7)$$

Where the two-by-two DD estimators are:

$$\hat{\beta}_{kU}^{2x2} \equiv (\bar{y}_k^{POST(k)} - \bar{y}_k^{PRE(k)}) - (\bar{y}_U^{POST(j)} - \bar{y}_U^{PRE(j)})$$

¹⁰ Note that the first two terms in equation (6) collapse to a function of pre/post mean differences because $D_{it} = \bar{D}_i = 0$ in the untreated group, $D_{it} = 0$ in pre-treatment periods, and $\bar{y}_i = \bar{D}_i \bar{y}_i^{POST(k)} - (1 - \bar{D}_i) \bar{y}_i^{PRE(k)}$. The sums over i thus include only the treated units and the sum over t includes only post-periods ($t > t_k^*$). Similarly, sum over time can be broken into pieces based on the different groups' treatment times, and the time means of y_{it} weight together each group's mean. $\widehat{var}(\tilde{D}_{it})$ follows from replacing y_{it} with D_{it} in the expression for $\widehat{cov}(\tilde{y}_{it}, \tilde{D}_{it})$.

$$\begin{aligned}\hat{\beta}_{k\ell}^{2x2,k} &\equiv \left(\bar{y}_k^{MID(k,\ell)} - \bar{y}_k^{PRE(k)}\right) - \left(\bar{y}_\ell^{MID(k,\ell)} - \bar{y}_\ell^{PRE(k)}\right) \\ \hat{\beta}_{k\ell}^{2x2,\ell} &\equiv \left(\bar{y}_\ell^{POST(\ell)} - \bar{y}_\ell^{MID(k,\ell)}\right) - \left(\bar{y}_k^{POST(\ell)} - \bar{y}_k^{MID(k,\ell)}\right)\end{aligned}$$

the weights are:

$$\begin{aligned}s_{kU} &= \frac{n_k n_U \bar{D}_k (1 - \bar{D}_k)}{\widehat{var}(\tilde{D}_{it})} \\ s_{k\ell} &= \frac{n_k n_\ell (\bar{D}_k - \bar{D}_\ell)(1 - (\bar{D}_k - \bar{D}_\ell))}{\widehat{var}(\tilde{D}_{it})} \\ \mu_{k\ell} &= \frac{1 - \bar{D}_k}{1 - (\bar{D}_k - \bar{D}_\ell)}\end{aligned}$$

and $\sum_{k \neq U} s_{kU} + \sum_{k \neq U} \sum_{\ell > k} s_{k\ell} = 1$.

Proof: See appendix A.

Theorem 1 completely describes the sources of identifying variation in a general DD model and their importance. With K timing groups, one could form $K^2 - K$ “timing-only” estimates that either compare an earlier- to a later-treated timing group ($\hat{\beta}_{k\ell}^{2x2,k}$) or a later- to earlier-treated timing group ($\hat{\beta}_{k\ell}^{2x2,\ell}$). With an untreated group, one could form K 2x2 DD’s that compare each timing group to the untreated group ($\hat{\beta}_{kU}^{2x2}$). Therefore, with K timing groups and one untreated group, the DD estimator comes from K^2 distinct 2x2 DDs.

The weights come both from group sizes, via the n_j ’s, and the treatment variance in each pair.¹¹ With one treated group, the variance of the treatment dummy is $\bar{D}_k(1 - \bar{D}_k)$, and is highest

¹¹ Many other least-squares estimators weight heterogeneity this way. A univariate regression coefficient equals an average of coefficients in mutually exclusive (and demeaned) subsamples weighted by size and the subsample x - variance:

$$\hat{\alpha} = \frac{\sum_i (y_i - \bar{y})(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2} = \frac{\sum_A (y - \bar{y})(x - \bar{x}) + \sum_B (y - \bar{y})(x - \bar{x})}{\sum_i (x - \bar{x})^2} = \frac{n_A s_{xy}^A + n_B s_{xy}^B}{s_{xx}^2} = \frac{n_A s_{xx}^{2,A}}{s_{xx}^2} \hat{\alpha}_A + \frac{n_B s_{xx}^{2,B}}{s_{xx}^2} \hat{\alpha}_B$$

Similarly, the Wald/IV theorem (Angrist 1988) shows that any IV estimate is a linear combination of Wald estimators that compare two values of the instrument. Gibbons, Serrato, and Urbancic (2018) show a nearly identical weighting formula for one-way fixed effects. Panel data provide another well-known example: a pooled regression coefficients equals a variance-weighted average of two distinct estimators that each use less information: the between estimator for subsample means, and the within estimator for deviations from subsample means.